

AMC Mark 1

Team: AMC

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1. Design Objective

The proposed design is a compact, two-layer STM32F401RE-based evaluation and development board inspired by the functionality and architecture of the STM32 NUCLEO-F401RE. The objective is to provide a practical development platform with the essential facilities required for programming, debugging, power management, communication, GPIO expansion, and user interaction.

The board uses the **STM32F401RET6** as the main microcontroller and integrates USB connectivity, regulated 3.3 V power, SWD programming/debugging, UART, SPI, I²C, GPIO expansion, user buttons, and status LEDs.

The design was developed independently using **KiCad**, with emphasis on compact component placement, reliable power distribution, organized routing, proper decoupling, and a clear PCB silkscreen.

The PCB was designed with a **70 mm × 60 mm board size**, which is well within the competition's maximum permitted size of **100 mm × 100 mm**. The design uses **exactly two copper layers (Top and Bottom)**, with both layers utilized for efficient signal routing and power/ground distribution. The compact 70 mm × 60 mm form factor was chosen to achieve efficient component placement while maintaining clear routing, adequate spacing, and good overall board organization.

2. Main Components and Features

The major components and functional blocks incorporated in the board are:

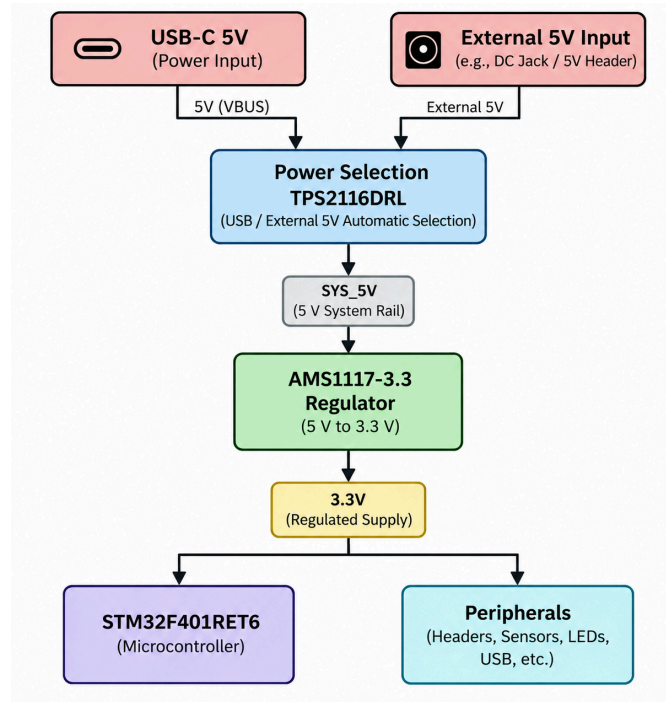
- **STM32F401RET6** – Main 32-bit ARM Cortex-M4 microcontroller
- **USB Type-C interface** – USB connectivity for the board
- **TPS2116DRL power multiplexer** – Power-source selection/management
- **AMS1117-3.3 regulator** – Generation of the regulated 3.3 V supply
- **TPD2EUSB30** – USB protection
- **SWD header** – Programming and debugging interface

- **UART header** – Serial communication
- **SPI header** – High-speed peripheral communication
- **I²C header** – Two-wire peripheral communication
- **GPIO expansion header** – Access to STM32 GPIO pins
- **Reset button** – Hardware reset functionality
- **User button** – General-purpose user input
- **User LED** – User/status indication
- **Power LED** – Power status indication
- **Crystal oscillator** – External clock source for the microcontroller
- **Decoupling capacitors** – Local power-supply stabilization
- **Protection components** – Input and USB protection

3. Power Architecture

The power architecture was designed to support both USB-C 5 V and an external 5 V input. The TPS2116DRL power multiplexer provides controlled power-source selection and supplies the SYS_5V rail. This is followed by the AMS1117-3.3 regulator, which generates the regulated 3.3 V supply for the STM32F401RET6 and associated peripherals. This architecture provides a flexible power-input arrangement while maintaining a regulated supply for the low-voltage circuitry.

The basic power flow is



The power architecture supports two 5 V input sources: USB-C 5 V and an external 5 V supply. These inputs are connected to the TPS2116DRL power multiplexer, which provides controlled selection and distribution of the available power source to the SYS_5V rail. The SYS_5V rail then feeds the AMS1117-3.3 regulator, which generates the regulated 3.3 V supply required by the STM32F401RET6 and associated low-voltage circuitry.

A dedicated power LED provides visual indication that the board is powered. The resulting architecture provides the essential functionality required for an STM32 evaluation and development platform while maintaining a compact PCB implementation.

4. PCB Design

The PCB was designed as a **two-layer board**, consisting of top and bottom copper layers. The maximum permitted board size for the competition is **100 mm × 100 mm**, and the board layout was developed within this constraint.

Component placement was carried out according to functional requirements and routing convenience. The STM32F401RET6 is positioned as the central processing element, while connectors and interfaces are arranged around the board to provide convenient access during development.

The major PCB design considerations were

- Compact component placement
- Proper power and ground distribution
- Short connections for critical power paths
- Appropriate placement of decoupling capacitors
- Organized signal routing
- Separation and organization of different functional interfaces
- Efficient use of both copper layers
- Accessible programming and communication headers
- Clear and readable PCB silkscreen

The final PCB passed the KiCad DRC with zero errors, with zero unrouted connections. The silkscreen was also organized to clearly identify important connectors, interfaces, power rails, and user controls.

5. Interfaces

5.1 SWD Interface

A dedicated **SWD (Serial Wire Debug)** header is provided for programming and debugging the STM32F401RET6. The interface provides the required connections.

- SWDIO
- SWCLK
- NRST
- VCC
- GND

The ST-Link circuitry itself is not included on the board, as it is not required by the competition specification. An external ST-Link programmer/debugger can therefore be connected through the SWD header.

5.2 UART Interface

A dedicated UART header is provided for serial communication with external devices. This interface can be used for debugging, terminal communication, data logging, and communication with other embedded systems.

5.3 SPI Interface

An SPI header provides access to the STM32 SPI peripheral signals. It can be used to interface external sensors, displays, memory devices, ADCs, DACs, and other SPI-compatible peripherals.

5.4 I²C Interface

An I²C header is provided for connecting multiple peripheral devices using the two-wire I²C communication protocol. This makes the board suitable for interfacing sensors, displays, EEPROMs, and other I²C devices.

5.5 USB Interface

A USB Type-C connector is provided as the primary USB interface. The USB section also includes dedicated protection circuitry to improve the robustness of the USB connection.

5.6 GPIO Expansion

A dedicated GPIO expansion header provides access to selected STM32 GPIO, power, and ground connections. This allows external circuits and modules to be connected easily during development and experimentation.

6. Design Decisions

Several design decisions were made to achieve a compact and functional evaluation board.

Power Management

The power architecture was designed to support both USB-C 5 V and an external 5 V input. The TPS2116DRL power multiplexer provides controlled power-source selection and supplies the SYS_5V rail. This is followed by the AMS1117-3.3 regulator, which generates the regulated 3.3 V supply for the STM32F401RET6 and associated peripherals. This architecture provides a flexible power-input arrangement while maintaining a regulated supply for the low-voltage circuitry.

Component Placement

The STM32F401RET6 was positioned as the central component of the PCB, with supporting power and clock components placed close to the MCU. Connectors were positioned around the board edges where possible to make external connections easier.

Interface Organization

Communication and programming interfaces were separated into dedicated headers for **SWD**, **UART**, **SPI**, and **I²C**. This makes the board easier to use and allows different peripherals or debugging equipment to be connected without interfering with other interfaces.

Routing Optimization

The PCB routing was planned to minimize unnecessary trace lengths and reduce routing complexity. Power connections and important signal paths were given appropriate routing consideration, while the two copper layers were utilized efficiently to complete the design within the board-size and layer constraints.

Ground and Power Distribution

Ground and power connections were distributed throughout the PCB with attention to maintaining reliable connections between the power supply, microcontroller, and peripheral

circuits. Decoupling capacitors were placed near relevant power pins to provide local supply stabilization.

Silkscreen and Usability

Clear silkscreen labels were provided for major connectors, interfaces, power rails, buttons, LEDs, and other important board features. This improves usability by allowing users to identify the purpose and pin functions of the different headers directly from the PCB.

7. Conclusion

The resulting STM32F401RE evaluation board provides a compact and organized development platform centered around the **STM32F401RET6** microcontroller. It integrates regulated power, USB connectivity, SWD debugging, UART, SPI, I²C, GPIO expansion, user controls, and status indicators in a two-layer PCB.

The design focuses on electrical completeness, practical component placement, organized routing, power integrity, and usability. The PCB was developed independently in KiCad while using the STM32 NUCLEO-F401RE as a functional reference, in accordance with the competition requirements. The final design is intended to provide a practical platform for STM32-based embedded development and peripheral interfacing.